Maintenance Schedule

Diesel Engine-Generator Sets
EPA
DG12V1600C6N
DG12V1600C7N
DG12V1600C8N
Stationary emergency only
Application group 3F

MSE50824/00E

A Rolls-Royce solution

System designation		Engine model	kW/cyl.	Application group
Product designation	Model number			
mtu 12V1600 DS750	DG12V1600C6N	12V1600G21S	63.3 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS800	DG12V1600C7N	12V1600G31S	67.4 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS900	DG12V1600C8N	12V1600G41S	75.4 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN

Table 1: Applicability

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Definition of Rolls-Royce Solutions

Rolls-Royce Solutions refers to Rolls-Royce Power Systems AG and its subsidiaries Rolls-Royce Solutions GmbH and Rolls-Royce Solutions America Inc.

Maintenance concept

The Maintenance Schedule applies to products which have been certified in accordance with the emissions regulations of the United States Environmental Protection Agency (U.S. EPA) and the California Air Resources Board (CARB).

The following instructions must be observed for these products:

The Maintenance Schedule is an integral part of the emissions certification. Failure to comply with the maintenance instructions may entail a violation of statutory emissions regulations mandated by the U.S. EPA and CARB. Modification or removal of mechanical or electronic components, or the installation of additional components, as well as the execution of calibration processes which might affect the emissions characteristics of the product are prohibited by the emissions regulations of the U.S. EPA and CARB.

Although Rolls-Royce Solutions' warranty obligations are not dependent upon the use of any particular brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only components approved by Rolls-Royce Solutions or their equivalent.

Emissions-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all component parts of the exhaust gas aftertreatment systems.

The use of spare parts which are not of equivalent quality to components approved by Rolls-Royce Solutions may impair the effectiveness of emissions control systems. The maintenance, replacement, or repair of the emissions control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

With the exception of those maintenance activities denoted as being crucial to emissions compliance in this Maintenance Schedule, the owner or operator of the product is not required to perform other maintenance activities listed in this Maintenance Schedule in order to uphold the emissions-related warranty. Rolls-Royce Solutions will not deny any emissions-related warranty claim based solely on the owner's or operator's failure to perform other recommended maintenance or non-essential emissions-related maintenance. However, Rolls-Royce Solutions may deny emissions-related warranty claims should the owner or operator have failed to perform maintenance denoted as being crucial emissions-related maintenance, or warranty claims relating to malfunctions in emissions-related components resulting from improper maintenance or usage by the owner or operator, incidents beyond the control of Rolls-Royce Solutions, or force majeure.

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience.

Observance of the specified maintenance intervals is essential to maintain product safety.

Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Maintenance activities are due either on reaching the specified number of operating hours, or on expiry of the time limit, whichever occurs first.

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The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the Maintenance Schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this Maintenance Schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Special conditions

Designation	Value
Load factor	≤ 100%
Operating hours	Unlimited hours
Time between major overhaul	8,000 h or 18 years

Table 2: Special conditions

Note 01:
The engines are certified for operation with the fuels approved in the Fluids and Lubricants Specifications. For operation with a high sulfur content in the fuel, the following must be observed: • The component TBO specified in the maintenance schedule refers to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590 or ULSD. If a fuel with a sulfur content > 15 ppm is used, the times specified in the maintenance schedule for component TBO of the cylinder head may be shorter. For reduced TBOs, refer to the Fluids and Lubricants Specifications. Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

Note 02:
If the engine is used in an UPS unit with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions

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1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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